

2D Transformations and Warping

Discussion #4

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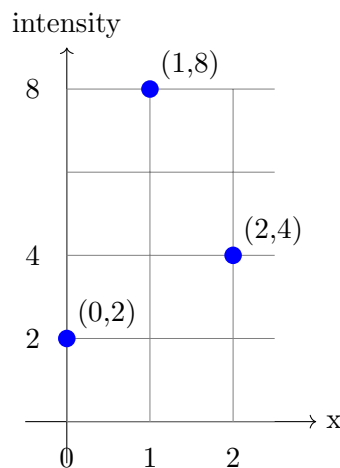
Topics

This section covers 2D transformations, interpolation, and image warping.

1 Interpolation

Problem 1.1: 1D Linear Interpolation.

Consider a 1D signal with 3 sample points at locations 0, 1, and 2 with values 2, 8, and 4 respectively:



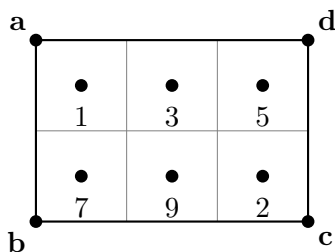
The signal is sampled at the following x locations using linear interpolation. What should the interpolated values be?

Location 0.0: _____

Location 0.5: _____

Location 1.75: _____

Problem 1.2: 2D Bilinear Interpolation. Consider a 2×3 grayscale image (2 rows, 3 columns) with the following pixel values (The numbers (1, 3, 5, 7, 9, 2) represent the pixel values at their respective centers):



Part (a): What are the (x, y) coordinates for points a, b, c, and d?

Part (b): The image is sampled at the following locations. What should the bilinearly interpolated values be?

Location (0.5, 0.5): _____

Location (0.5, 0.0): _____

Location (1.5, 1.0): _____

2 Warping

Problem 2.1: *Reverse Mapping with Inverse Transform*

We have a 2×3 source image with the following pixel values:

• 1	• 2	• 3
• 4	• 5	• 6

We want to apply transformation $T = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ (scale by 2).

Part (a): Find the inverse transformation T^{-1} .

$$T^{-1} = \begin{bmatrix} _ & _ \\ _ & _ \end{bmatrix}$$

Part (b): For output pixel (3,1), use T^{-1} to find which source pixel to sample from.

$$T^{-1} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \underline{\hspace{2cm}}$$

Source coordinates: _____

Pixel value: _____

Part (c): For output pixel (2,2), use T^{-1} to find which source pixel to sample from.

$$T^{-1} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \underline{\hspace{2cm}}$$

Source coordinates: _____

Part (d): For (c), show how to use bilinear interpolation to get the pixel value.

Which 4 source pixels are involved? _____

What are their values? _____

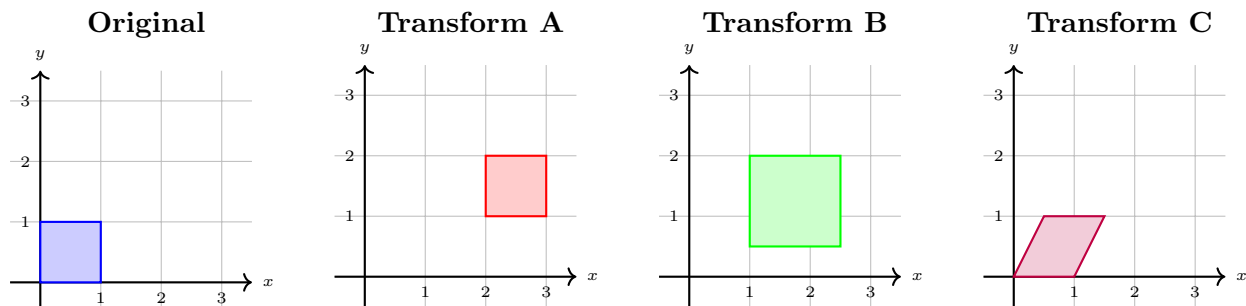
Interpolated value: _____

Part (e): Why do we need the inverse transformation for reverse mapping?

3 Transformation Types

Problem 3.1: *Match that transformation*

Consider the original square with vertices at (0,0), (1,0), (1,1), and (0,1). Below are three transformations applied to this square:



Match each transformation above with its corresponding transformation matrix:

Matrix I: $\begin{bmatrix} 1.5 & 0 & 1 \\ 0 & 1.5 & 0.5 \\ 0 & 0 & 1 \end{bmatrix}$ matches transformation _____

Matrix II: $\begin{bmatrix} 1 & 0.5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ matches transformation _____

Matrix III: $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ matches transformation _____

Problem 3.2: Homogeneous coordinates

Suppose we want to translate an image by $(2, 3)$.

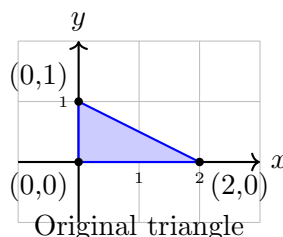
Part (a): Why can't we represent translation $(x, y) \rightarrow (x + 2, y + 3)$ as a 2×2 matrix?

Part (b): Write the 3×3 homogeneous matrix for translating by $(2, 3)$.

$$T = \begin{bmatrix} _ & _ & _ \\ _ & _ & _ \\ _ & _ & _ \end{bmatrix}$$

Part (c): Consider the triangular object with vertices at $(0,0)$, $(2,0)$, and $(0,1)$. Compare what happens when we apply transformations TR versus RT , where T is translation by $(3,0)$ and R is 90° counterclockwise rotation.

$$\text{Given: } T = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ and } R = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

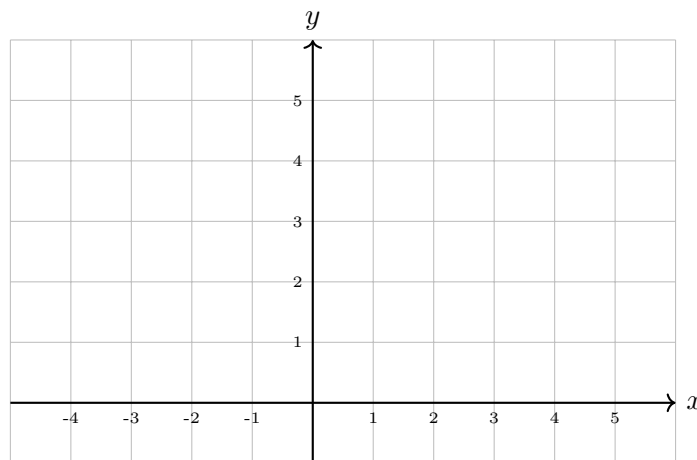


Calculate the composite transformations:

$$TR = \begin{bmatrix} _ & _ & _ \\ _ & _ & _ \\ _ & _ & _ \end{bmatrix} \quad \text{New vertices: } \underline{\hspace{10cm}}$$

$$RT = \begin{bmatrix} _ & _ & _ \\ _ & _ & _ \\ _ & _ & _ \end{bmatrix} \quad \text{New vertices: } \underline{\hspace{10cm}}$$

Draw both transformed triangles on the graph below:



Draw TR triangle and RT triangle